



First pop quiz!

What do you expect from this course?

Happy New Year!



Let's decompose the title!

Software Design I - Introduction to Software Development

- **Software Design:** (PEng core skill)
 - *"the process of defining the architecture, components, interfaces, and other characteristics of a system or component"* (source: SWEBOX)
- **Introduction:**
 - You've never done that before. (source: me, chair of the SE committee for CAS)
- **Software Development:**
 - *"A set of computer science activities dedicated to the process of creating, designing, deploying and supporting software"* (source: IBM)

**So this is absolutely
not about coding!**

(coding skills are essential, and assumed now at Level II)

**We now move on to the
Engineering dimension.**

(let the fun begin!)

Quotes from alumni

- From the **student discord server**:
 - "That m'th'f'ck'r piece of sh't is on an ego trip and just here to **destroy your mental health**. His assignments are impossible and **ridiculous** if you are not working 100% of your time on these stupid things that are totally **useless**."
- From **student discussions and town halls**:
 - "This course gave me **all the skills I need to land a dream coop internship**. It was a **game changer** during the technical interview process."
- From **MS Teams direct message**:
 - "I have to apologize: **I thought your course was a scam**. Then I went into coop during summer, and I realized I should have listened to the contents of the course rather than fighting it. **I lost all my summer evenings and weekends, re-learning by myself everything I should have learned during 2AA4** to catch up with my team at work."



Team & Logistics

Who are we?



The Amazing 2AA4 Team



**Alexandre
Lachance**

Q&A, Support



**Mohamed
Aly**

Tutorial T02



**Richard
Li**

Tutorial T02



**Eshaan
Chaudhari**

Tutorial T01



Yours Truly

Instructor



Sébastien Mosser

Snowboarding since 1995 Composing things since 2007



- 22-....: Associate Professor, McMaster Univ.
- 19-21: "Professeur", UQAM
- 12-18: "Maître de Conférences", Univ. Côte d'Azur
- 11-12: Research Scientist, SINTEF
- 10-11: Postdoc, Inria Lille Nord-Europe
- 07-10: PhD student, Université de Nice



It takes a village to build a course



Dr. Jean Privat
(UQAM)



Dr. Anne-Marie Dery
(UCA)



Dr. Xavier Blanc
(UBordeaux)



Dr. Philippe Collet
(UCA)



Dr. Perdita Stevens
(Univ. Of Edinburgh)



Dr. Jean-Michel Bruel
(Univ. Of Toulouse)



Dr. Spencer Smith
(McMaster Univ.)



Dr. Mireille Blay-Fornarino
(UCA)



Dr. Martin Robillard
(McGill)



Dr. Richard Paige
(McMaster Univ.)

Logistics: Lectures & Tutorials

Slot	Monday	Tuesday	Wednesday	Thursday	Friday
08:30 10:20					T03
12:30 13:20		Lecture #1	Lecture #2		Lecture #3
14:30 16:20				T01 & T02	

Lectures and **Tutorials** are here to help with your assignments

Office hours: appointment only (send DM on MS Teams)

Emails will NEVER be answered

(damned Office365 quarantine & stupid spam filters)

We ONLY use MS Teams as a dedicated communication channel for 2AA4.

MS Teams → like internal Stack Overflow
Do not use screenshots, rather copy-paste text

Logistics: Assignments

ID	Assignment	Kind	Weight	Start date	Delivery date
A1	Coding & Releasing	Individual	10%	09/01/24	28/01/24
A2	Group Project	Team	30%	30/01/24	24/03/24
A3	Evolution of legacy code	Individual	10%	26/03/24	10/04/24
MT	Midterm exam	Individual	15%	27/02/24	
EX	Final exam (2.5 hours)	Individual	35%	To be announced (registrat)	

Improve A1

Group work is 30% of your grade. Individual work is 70%.

Support: Teams to interact on dedicated channels (only)

Not asked to write code on midterm/final
Rather REFLECTING on code will be in midterm/final

2A4 requires continuous effort



Student working regularly on the assignments



Instructor when grading an assignment that was rushed two hours before the deadline and has absolutely no depth of thoughts.

MSAF Policies, SAS & RISO accommodations

- **Type-A (self-reported) MSAFs** are **automatically** accepted.
 - *No need to send an email. I'll not answer.*
- For **assignments A1 and A3**: Three extra days are added to the deadline.
- For the **midterm**: The weight carries on to the final (35% → 50%)
- **Type-B (documented) MSAFs** are discussed on a case-by-case basis
 - The objective is to find the **best way to accommodate your situation**.
- **SAS accommodations** need to be processed through SAS. It takes time.
- **RISO accommodations** (exams) also have tight deadlines.

Academic dishonesty

The department is enforcing a
zero-tolerance policy for
cheating and **plagiarism**.

Course Contents

What are we covering in this course?

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Prerequisites for 2AA4

- **SFWRENG 2DM3: *Discrete Mathematics with Applications I***
 - Set theory, operators, relations (equivalence/identity), substitutions
- **SFWRENG 2OP3: *Object-oriented programming***
 - Classes, interfaces, visibility, message sending, static & instance
 - Java programming ecosystem
- **SFWRENG 2XC3: *Practice & Experience - Development Basics***
 - Unix/Linux environment, scripting, version control (T01)

We will also assume you know how to maintain a healthy development environment on your computer.

2AA4 in a nutshell



Topics are organized "by week"

1. From Coding to Development
2. Organizing your code
3. Building the product right
4. Contracts & Responsibilities
5. Building the right product
6. Algebraic foundations
7. Designing with polymorphism
8. Case study: MacUnit
9. The need for patterns
10. Patterns in action
11. Case study: Pure OO design
12. Metrics & Visualization

Tutorials complement the lectures

- | | |
|--|--|
| 1. <i>From Coding to Development</i> | (A) Engineering Software |
| 2. <i>Organizing your code</i> | (B) Vertical Feature Decomposition |
| 3. <i>Building the product right</i> | (C) Module Interface Specification |
| 4. <i>Contracts & Responsibilities</i> | (D) Modelling with the UML (structure) |
| 5. <i>Building the right product</i> | (E) Unit Testing |
| 6. <i>Algebraic foundations</i> | (F) Modelling with the UML (behaviour) |
| 7. <i>Designing with polymorphism</i> | (G) Code Quality |
| 8. <i>Case study: MacUnit</i> | (H) Case study: Library design |
| 9. <i>The need for patterns</i> | (I) Case study: Fancy Coffee Maker |
| 10. <i>Patterns in action</i> | (J) Case study: Arithmetic expressions |
| 11. <i>Case study: Pure OO design</i> | (K) Analysis of Patterns |
| 12. <i>Metrics & Visualization</i> | (L) Case study: Poker hands |

**Tutorials covers material that is
not 100% covered in the lectures**
(and yes, "it'll be at the exam")

**Tutorials are designed to
advance your assignments and
help you prepare for exams.**

Assignments relies on tutorials & lectures

1. From Coding to Development	A1	(A) Engineering Software
2. Organizing your code		(B) Vertical Feature Decomposition
3. Building the product right		(C) Module Interface Specification
4. Contracts & Responsibilities	A2	(D) Modelling with the UML (structure)
5. Building the right product		(E) Unit Testing
6. Algebraic foundations		(F) Modelling with the UML (behaviour)
7. Designing with polymorphism		(G) Code Quality
8. Case study: MacUnit		(H) Case study: Library design
9. The need for patterns	A3	(I) Case study: Fancy Coffee Maker
10. Patterns in action		(J) Case study: Arithmetic expressions
11. Case study: Pure OO design		(K) Analysis of Patterns
12. Metrics & Visualization		(L) Case study: Poker hands

- Final will be a case study
- Ask for help when needed

This course is a transformative one



Transformation $\Rightarrow$ discomfort

You **poor unfortunate soul**
It's sad but true
If you want to cross the bridge my sweet
You have to pay the toll
Take a gulp and take a breath
And go ahead and sign the scroll

- Ursula, The Sea Witch



Bibliography & Textbooks

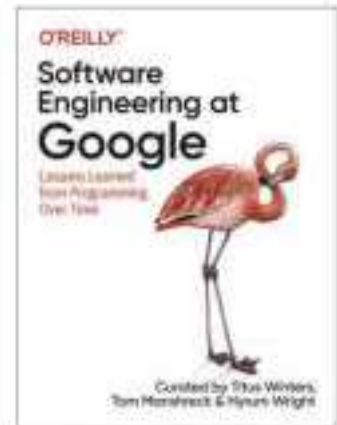
(all free, see links on Avenue)



(Robillard, 2022)



(Martin, 2008)



(Winters et al, 2021)

How to fail pass 2AA4?

(Works with other courses too by the way)



How to succeed easily?

**Practice
Practice
Practice**

How do we support you?

- **We are here to help** you succeed (*unbelievable... isn't it?*)
 - **Lectures, tutorials** and **assignments** are carefully tailored to be aligned
 - **2AA4 is synchronized with 2CO3** in terms of theoretical knowledge
- **"I'm fed up with these vague assignments and imprecise tasks."**
 - This is the **very essence** of **software engineering**.
 - **You're here to learn** how to deal with imprecise demands and vague stuff!
 - Two easy ways to deal with it: (1) **anticipate**, and (2) **ask questions!**
- **MS Teams is here to help**. Ask questions on the right channels, we'll help!

Opportunities for extra-credit.

- **Meaningful & Memorable (M&M)**
 - Each week, a **survey on Avenue** to help you **reflect on your learning process**
 - Also used as a **barometer for your progress**
 - Submit your M&M **after the last lecture of the week**
 - We'll grade your reflections at the end of the semester (**up to +1 on final grade**)
- **Course Evaluation Survey**
 - Student participation is historically low in the department. 🙄
 - **Numbers of answers** will "pimp" the **grade rounding** method (see syllabus)

Conclusions

That being said, ... It'll be a lot of fun

Deliver “on time, on specs, and on budget”

- Transformative course
 - Your first “Software Design” course ever!
- Four credits course
 - So expect 30% of additional work compared to a three credits one.
- From *programmer* to *developer*:
 - Product development
 - Object-oriented design
 - Software quality
- Cool assignments!



(* That's not true, assuming you work regularly)

Tutorials start this week!



ENGINEERING

Computing
& Software

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ITB 131 (appointment only)

